Ch. 15 HW #2 – Solutions

Principles of Microeconomics

December 10, 2025

Quick Quiz

4. (d) – Profit maximizing firms produce the quantity at which marginal revenue (price) equals marginal cost, so the marginal cost curve determines the quantity supplied for a given price. In the short run, firms' fixed costs are lost regardless of whether they produce, so they will produce as long as their revenue is greater than their variable costs.

Problems & Applications

6. In a competitive market, all of a firm's output gets sold at the market price. That means marginal revenue is equal to average revenue which is equal to the market price of the good, Therefore, the average revenue and the price of the good both equal \$10. Since the price is \$10, that means the firm sold \$500/\$10 = \$50

7. (a) Price equals average revenue which equals \$10. Therefore, profit is

$$\pi = (P - ATC) * Q = (10 - 8) * 100 = 200.$$

   (b) Since the firm is profit maximizing, marginal cost equals marginal revenue which equals the price of the good, \$10.

   (c) Average variable cost is

$$AVC = ATC - AFC = ATC - FC/Q = 8 - 200/100 = \$6.$$

   (d) Since marginal cost is greater than average total cost, average total cost is greater than its minimum. That means the firm's efficient scale is less than 100 units.